SMART INDIA HACKATHON 2026

HARDWARE EDITION | POWERED BY ARDUINO UNO Q (QUALCOMM INC.)

PROJECT PHOENIX X

SAR UAV

Arduino UNO Q Dual-Brain Edge-AI Scout for Zero-Infrastructure Disaster Reconnaissance

PROBLEM STATEMENT ID

SIH26177

THEME

Robotics and Drones

PROBLEM STATEMENT TITLE

A deployable AI-powered autonomous drone that aids search-and-rescue operations by detecting people and hazards, thereby improving responder safety and reducing victim discovery time.

SPONSORING ORGANIZATION

Qualcomm Inc.

CATEGORY

Hardware

TEAM NAME

Phoenix X

TEAM ID

141943

WEB DEMO

phoenix-sar-uav.vercel.app

VIDEO

https://youtu.be/NXkJmau0QLo?t=148



- Arduino UNO Q Dual-Brain SBC
- QRB2210 Linux + STM32U5 RTOS
- Dual 1080p + Thermal Fusion
- 3-5 km LoRa Telemetry
- IEEE Anti-Drift VIO Navigation

PROJECT PHOENIX

UAV for Zero-Infrastructure Survivor Geolocation & Resilient LoRa Mesh

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PROBLEM STATEMENT

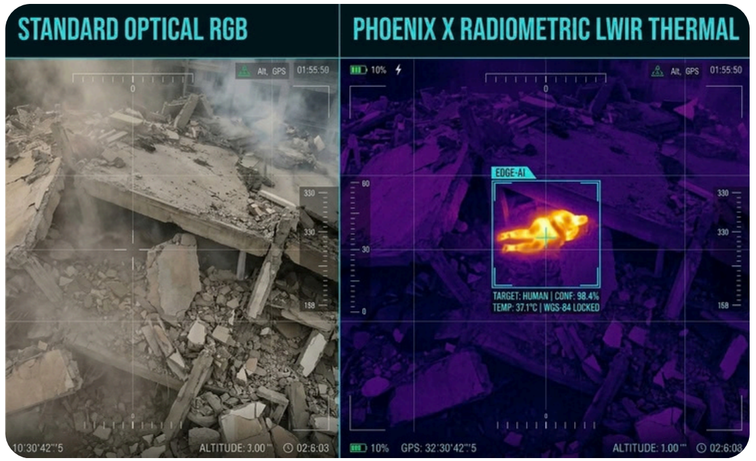
CRITICAL DISASTER BARRIERS — FOUR OPERATIONAL FAILURES:

01 — TOTAL COMMS BLACKOUT: Cellular and Wi-Fi networks down; cloud-reliant commercial drones fail immediately.

02 — ZERO OPTICAL VISIBILITY: Darkness, dense smoke, and debris dust blind conventional visible-light RGB cameras.

03 — UNLOCALIZED CASUALTIES: Visual spotting lacks coordinates; rescue teams dig blindly without actionable GPS fixes.

04 — SECONDARY RESCUER HAZARDS: Live electrical lines, shifting rubble piles, and gas voids endanger ground rescue crews.



PROPOSED SOLUTION

PROJECT PHOENIX: ARDUINO UNO Q DUAL-BRAIN ARCHITECTURE

ARDUINO UNO Q DUAL-BRAIN: QRB2210 MPU performs INT8 edge-AI on Linux; STM32U585 MCU runs deterministic 250 Hz flight loop on Zephyr.

1080P + THERMAL FUSION: Fuses 1080p optical with 8–14µm LWIR to detect human heat signatures in darkness, smoke & debris.

CONTINUOUS GEOREFERENCING: Synchronized GNSS, attitude, camera geometry & ToF altitude georeference victims in real time without hovering.

OFFLINE RESCUE ROUTING: Prioritizes triage on-chip; broadcasts 17-byte LoRa packets to chart Dijkstra corridors avoiding hazards [1, 3].

INNOVATION / UNIQUENESS

Dual-Brain Architecture

ARDUINO UNO Q SBC (4GB/32GB): Qualcomm Dragonwing QRB2210 MPU + STM32U585 MCU on a single 42g board.

5W Edge YOLO

SUB-5W HETEROGENEOUS AI: QRB2210 CPU/GPU/DSP executes INT8 YOLO inference, preserving battery for flight.

Ray-Cast Geolocation

REAL-TIME GEOREFERENCING: Tait-Bryan ray projection calculates ground WGS-84 coordinates on the fly without hovering.

ROS-LoRa P2P Bridge

17-BYTE LORA TELEMETRY: Off-grid bit-packed triage stream reaches 3+ km in complete cellular blackouts [3].

Anti-Compass Drift VIO

ANTI-DRIFT VIO NAVIGATION: Visual-Inertial Odometry overrides compromised magnetometers in metallic rubble environments.

TRADITIONAL DISASTER SEARCH (Current Approach)

Manual Foot Sweeps › Blind Rubble Digging › Stalled at Sunset (Darkness) › High Responder Casualty Risk › Critical Hours Lost

PHOENIX AUTONOMOUS SAR (Our Approach)

Autonomous Aerial Sweep › 24/7 Thermal Night Vision › Instant Survivor GPS Raycasting › Zero-Drop LoRa Telemetry › Safe Responder Path Routing

TECHNICAL APPROACH

Dual-Brain Architecture & Workflow

IMPLEMENTATION WORKFLOW

1. **Sector Partition & Launch**
   OFFLINE SECTOR SETUP: Operator defines 2 km² search polygon on tactical tablet; autonomous launch in < 3 minutes.

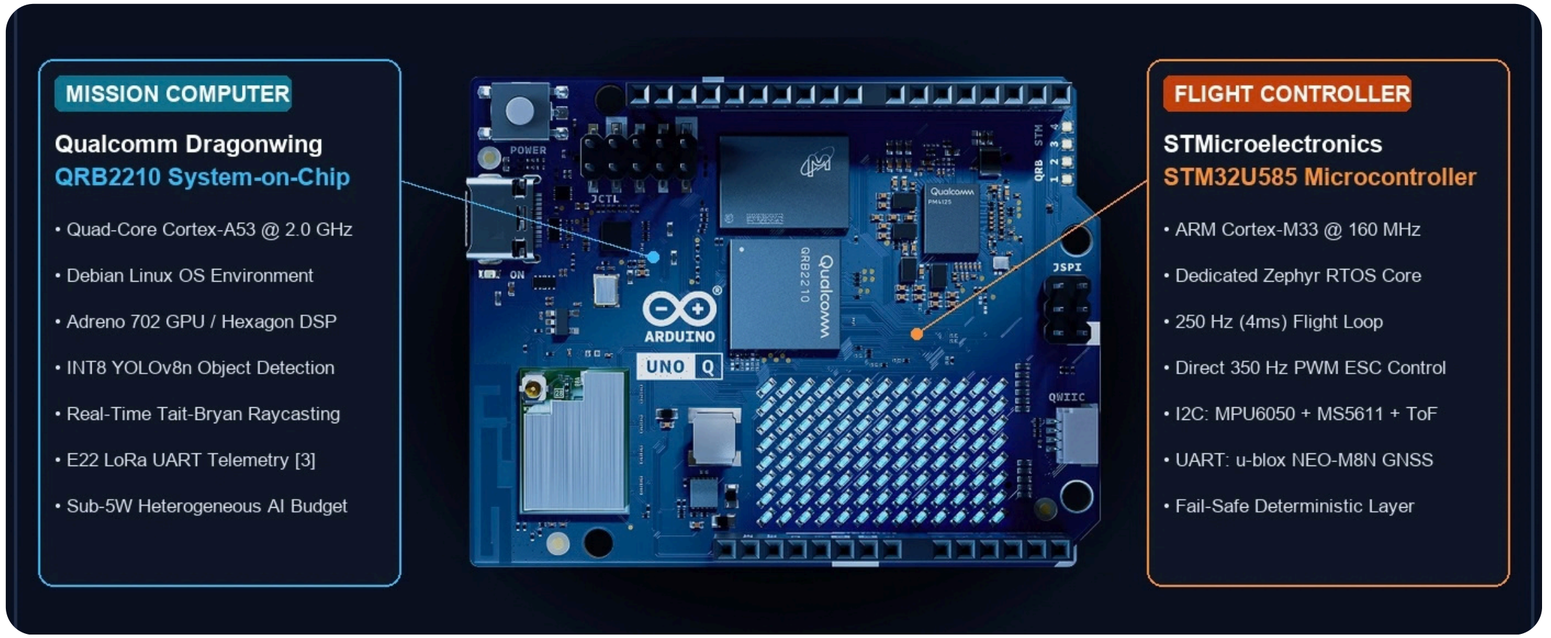
2. **Autonomous Grid Sweep**
   TERRAIN-FOLLOWING GRID: Boustrophedon sweep with 60% sidelap; downward ToF altimeter maintains constant GSD over sloped terrain.

3. **Multi-Spectral Perception**
   1080P + THERMAL FUSION: Dual 1080p RGB + LWIR thermal fusion; edge YOLO detects survivor heat anomalies in darkness, smoke & debris.

4. **Ray-Cast Geolocation**
   INSTANT GEOREFERENCING: Synchronized GNSS, attitude, camera geometry & ToF altitude georeference victims on the fly without hovering.

5. **Binary LoRa Packet Burst**
   17-BYTE LORA BURST: Onboard edge triage scores urgency and broadcasts compact binary frame over 868 MHz UART LoRa [3].

6. **GCS Triage & Route Planning**
   DYNAMIC SAFE ROUTING: Tactical GCS runs Dijkstra graph search [1] to generate hazard-free ground ingress corridors for first responders.





TECH STACK

Flight Stack	Zephyr RTOS, YMFC-32 Core, PX4 Autopilot
Edge AI & Vision	YOLOv8n INT8 (Detection & Avoidance), SNPE SDK, ONNX, OpenCV
Robotics Middleware	ROS2 Humble, MAVLink v2, micro-ROS
Long-Range RF Comms	Ebyte E22-900T22S LoRa (868 MHz UART)
Tactical GCS	React, Leaflet, Tailwind CSS, WebSockets
Security	HMAC-SHA256 Telemetry Packet Signatures

FEASIBILITY AND VIABILITY

Feasibility, Challenges-to-Strategies & Market Viability

FEASIBILITY

TECHNICAL FEASIBILITY

- **THRUST HEADROOM:** 2.23:1 maximum thrust-to-weight ratio provides substantial control authority in wind gusts.
- **SUB-5W COMPUTE:** Arduino UNO Q heterogeneous AI consumes <4% of power, reserving energy for rotor flight.

OPERATIONAL FEASIBILITY

- **RAPID SETUP:** < 3 minutes from rugged backpack unboxing to active search flight.
- **CONTINUOUS SORTIES:** Dual-pack quick-swap batteries support rolling 100% duty-cycle rescue sweeps.

FINANCIAL FEASIBILITY

- **PROTOTYPE BUILD (₹37,400):** Genuine Arduino UNO Q 4GB/32GB build for full hardware evaluation.
- **PRODUCTION BOM (₹1.2 LAKH):** Enterprise-grade SAR UAV at 1/10th the cost of imported commercial drones.

VIABILITY

ECONOMIC VIABILITY

Prototype cost: <40,000₹

Enterprise Drone Disruption costs:

DJI Matrice 30T **₹10+ Lakhs**

Phoenix **₹1.2 Lakh**

- **FLEET SCALABILITY:** NDRF battalions can deploy a 10-drone swarm for the acquisition cost of 1 imported drone.

AIR-GAPPED TACTICAL RESILIENCE

- **100% OFFLINE DATA PRIVACY:** Zero cloud dependencies; all sensor feeds and victim GPS fixes remain strictly local with first responders.

DEPLOYMENT ECOSYSTEM

- **FIELD-READY INTEGRATION:** Native protocol compatibility with NDRF, SDRF, and civil disaster response command units.

CHALLENGES → STRATEGIES

1. **Comms Blackout in Canyons**

   868 MHZ LORA BRIDGE: Sustains NLoS micro-telemetry packet link [2] without cellular towers.

2. **Smoke, Mud Slush & Darkness**

   LWIR THERMAL DETECTION: 8–14µm radiometric array isolates 37°C casualties through smoke, dust, and debris [1] .

3. **Rubble Steel Magnetometer Distortion**

   ANTI-DRIFT VIO NAVIGATION: Visual-Inertial Odometry overrides compromised magnetometers in collapsed steel rubble.

4. **Propeller Vibration Sensor Noise**

   HARMONIC NOTCH FILTERING: Silicone damping mounts + software notch filtering eliminate ESC/motor noise on 250Hz EKF loop.

5. **Actuator Saturation in Gusts**

   ANTI-WINDUP PID CLAMPING: Dynamic integrator clamping prevents control surface saturation during aggressive gusts.

IMPACT AND BENEFITS

Mission Impact & Multi-Stakeholder Benefits

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IMPACT PIPELINE

	1 DEPLOY	2 SEARCH	3 LOCATE	4 RESPOND	5 OUTCOME
PHOENIX DOES	**BACKPACK DEPLOY:** Unbox to autonomous flight in < 3 minutes	**BOUSTROPHEDON SWEEP:** Autonomous dual 1080p + thermal grid coverage	**INSTANT GEOREFERENCING:** Tait-Bryan projection calculates exact ground GPS	**TRIAGE & HAZARD ROUTE:** Ranks casualties & computes Dijkstra safe corridors [1]	**FLEET ECONOMICS:** Full autonomous capability at 1/10th enterprise cost
IMPACT	**ZERO INFRASTRUCTURE:** Search begins immediately upon arrival	**ALL-WEATHER SCAN:** Continuous search in smoke, dust & darkness	**ZERO SEARCH DELAY:** Rescuers proceed straight to survivor coordinates	**CREW PROTECTION:** Rescuers avoid live power cables & gas hazards	**LIVES SAVED:** Accelerates extraction within the Golden 72 hours
MEASURE	**< 3 min** unboxing to flight	**2 km²** search sector	**0 cell towers** needed to report locations	**Ranked queue** + safe rescue route	**₹1.2 Lakh** vs ₹12-14 Lakhs

TRAPPED VICTIMS

- **GOLDEN 72H WINDOW:** Rapid discovery within critical survival timeframe
- **DEBRIS & SMOKE SENSING:** Thermal lock in zero-light and smoky collapse voids
- **TRIAGE PRIORITIZATION:** Instant medical urgency rating streamed to rescue teams
- **SURVIVAL MULTIPLIER:** Dramatically cuts time-to-extraction under rubble

NDRF / SDRF TEAMS

- **ADVANCE HAZARD MAPPING:** Flags live high-voltage cables, gas leaks & floodwaters
- **SAFE ROUTE NAVIGATION:** Dynamic Dijkstra corridor guides rescuers around hazards [1]
- **TACTICAL AWARENESS:** Crews proceed straight to verified survivor coordinates
- **FATIGUE REDUCTION:** Replaces blind shovel digs with targeted extraction

DISASTER AUTHORITIES

- **UNIFIED COMMON PICTURE:** Real-time casualty mapping with micro-telemetry tracking
- **DYNAMIC SECTOR LOGS:** Autonomous coverage tracking ensures no sector is missed
- **17-BYTE LORA STREAM:** Ultra-compact bit-packed alerts arrive in under 3 seconds [3]

DISTRICT GOVERNANCE

- **TARGETED RESOURCE DROPS:** Prioritized airlift and medical kits based on triage data
- **AFFORDABLE PROCUREMENT:** Deploy 10 Phoenix drones for the price of 1 enterprise UAV
- **SCALABLE STOCKPILING:** Standardized COTS electronics enable rapid state-wide rollout

RESEARCH AND REFERENCES

Academic Research Foundations & Prototype Verification

RESEARCH & REFERENCES

01 **IIT Jodhpur & Manipal Univ. (IEEE Access, Sept 2026) [1]**
- *"SwarmSAR-FW: Drone-Based Swarm Search & Rescue in Fire and Water Disasters"*
- **HAZARD-AWARE DIJKSTRA ROUTING:** Validates safe ingress routing (98.7% triage accuracy, 4.1s) avoiding obstacles. | **DOI:** 10.1109/ACCESS.2026.3730315

02 **UPC Barcelona (IEEE Access, Sept 2026) [2]**
- *"Long-Range BLE Localization via Drone LoRa Relays for SAR Missions"*
- **868 MHZ LORA NLoS CANYON RELAY:** Validates off-grid BLE-to-LoRa payload forwarding across deep terrain. | **DOI:** 10.1109/ACCESS.2026.3731317

03 **Univ. of Cyprus & KIOS CoE (Elsevier Computer Communications, 2024) [3]**
- *"Mission-Critical ROS-LoRa Architecture for Autonomous Drone Search & Rescue"*
- **17-BYTE COMPACT TELEMETRY:** Validates ultra-low-bandwidth bit-packed triage frames over off-grid LoRa links. | **DOI:** 10.1016/j.comcom.2024.04.015

04 **TalTech Estonia (IEEE Access, July 2023) [4]**
- *"Applying RIS-Based Communication for Collaborative Swarm Computing"*
- **DECENTRALIZED EDGE COMPUTE:** Validates autonomous on-device compute (-32% energy, -18% latency) with zero cloud. | **DOI:** 10.1109/ACCESS.2023.3293737

05 **IEEE Trans. Geoscience & Remote Sensing (IEEE TGRS, Nov 2023) [5]**
- *"RGB-T Multimodal Salient Object Detection for Disaster Search and Rescue"*
- **DUAL-SPECTRAL AI FUSION:** Validates 1080p RGB + LWIR thermal fusion with cross-attention for zero-light triage. | **DOI:** 10.1109/TGRS.2023.3332179

06 **IEEE Trans. Geoscience & Remote Sensing (IEEE TGRS, Dec 2023) [6]**
- *"Robust Monocular Visual-Inertial Odometry in GNSS-Denied Disaster Rubble"*
- **GNSS-DENIED VIO SLAM:** Validates anti-drift monocular VIO state estimation & obstacle avoidance in rubble. | **DOI:** 10.1109/TGRS.2023.3323519

PROTOTYPE PROOF

OPEN-SOURCE REPOSITORY & VERIFICATION ASSETS

Group Presentation Video:
https://youtu.be/NXkJmau0QLo?t=148

Web Demo Video:
https://youtu.be/NXkJmau0QLo

Web Demo:
phoenix-sar-uav.vercel.app

Github Repo:
github.com/jvstin47/Phoenix-SIH26177-SAR-UAV

Documented Subsystems:

1. docs/research_papers_synthesis_2026.md
2. docs/prototype_bom_and_financial_viability.md
3. docs/ros_lora_mission_critical_uav_swarm_research.md
4. docs/hardware_viva_defense_guide.md

Prototype Architecture Deliverables:

- Live Web GCS Tactical Dashboard (React + Leaflet)
- Multi-Spectral Dual Stream (RGB + LWIR Thermal HUD)
- LoRa 17-Byte Binary Bit-Packing Serialization Module
- Ray-Casting Geolocation Python Engine